

Final Exam Exercise Problems

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1. Number Systems and Binary Representation

a) Change the numbers:

- i) 401_8 to 12-bit binary number system. Show the steps.
- ii) 53_{10} to 12-bit binary number system. Show the steps.

b) What is the representation of the following numbers in:

- i) 8-bit sign-and-magnitude
- ii) 8-bit two's complement

a) $+134_8$

b) -89_{10}

c) 00011010_2 and 11101110_2 are two integers in one's complement. Show the arithmetic operation to add these two integers.

d) Show the calculation for the operation $28 - 10$ in 8-bit two's complement representation.

e) Change the numbers:

i) 157_8

ii) 101_{10}

iii) $3C6_{16}$

to 10-bit binary number system. Show the steps.

f) What are the representations of:

i) $+144_8$

ii) -98_{10}

in:

a) 8-bit sign-and-magnitude

b) 8-bit two's complement

g) Show the calculation for $-2 - (-10)$ in one's complement (8-bit).

h) Show the calculation for $18 - 13$ in two's complement (8-bit).

2. Logic Circuits, Truth Tables, and Karnaugh Maps

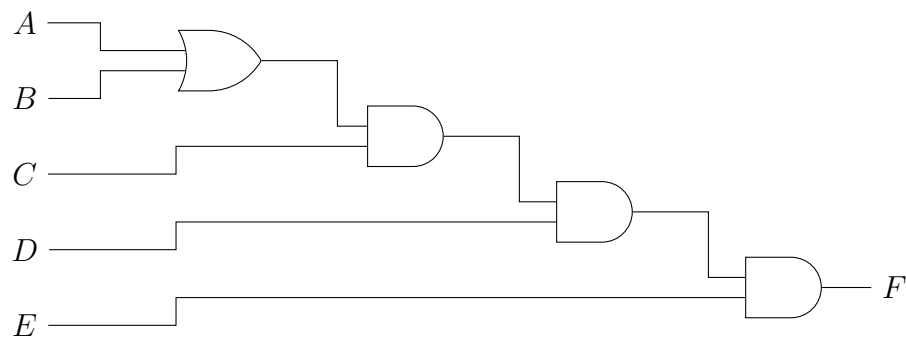
a) Draw the logical circuit for the following logical expression:

$$F = A + BC + \overline{D}$$

b) Draw the logical circuit for the following logical expression:

$$Q = (AB) + (\overline{A+B})$$

c) Draw the logic circuit for:



d) Draw a logic circuit for: $G = A\overline{B} + A\overline{C}$

e) Draw the circuit for (d) using ONLY NAND gates.

f) Provide logical expression for the logical table below.

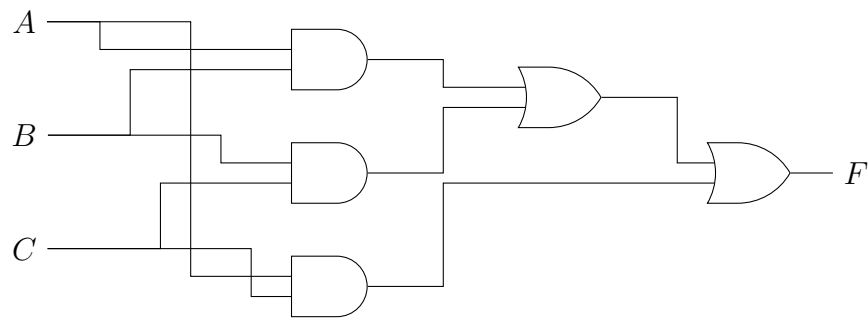
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

g) Create the truth table for: $F = X(Y + Z) + YZ$

h) From the truth table, develop the Karnaugh map and provide the final F.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

i) Given the logic circuit, determine output:



j) From Karnaugh maps, derive function F :

i) Left hand corner entries

	BC=00	BC=10	BC=11	BC=01
A=0	1	1	0	0
A=1	0	0	0	0

ii) All first row entries

	BC=00	BC=10	BC=11	BC=01
A=0	1	1	1	1
A=1	0	0	0	0

iii) Middle-column entries

	BC=00	BC=10	BC=11	BC=01
A=0	0	1	1	0
A=1	0	1	1	0

iv) All entries except left-bottom corner

	BC=00	BC=10	BC=11	BC=01
A=0	1	1	1	1
A=1	0	0	1	1

v) All entries except middle bottom

	BC=00	BC=10	BC=11	BC=01
A=0	1	1	1	1
A=1	1	0	0	1

vi) Provide logical expression for the logical table below.

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

3. Assembly Language Analysis

a) What is the final value in register CH?

```
A    DB    00110101B    ; 0x35
B    DB    00001111B    ; 0x0F

MOV   CX, 0000H
MOV   CL, A
SUB   CL, B
CMP   CL, 10H
JNE   L1
MOV   CH, B
L1:   ADD   CH, A
```

b) What is the final value in register AH?

```
P    DB    10011001B    ; 0x99
Q    DB    00010001B    ; 0x11

MOV   AX, 0000H
MOV   AL, P
ADD   AL, Q
CMP   AL, 0A0H
JE    L2
MOV   AH, Q
L2:   SUB   AH, P
```

c) What are the final values of AL and AH?

```
MOV   AL, 05H
MOV   AH, 02H
ADD   AL, AH

MOV   BL, 07H
DIV   BL
```

d) What are the final values of AX and DX?

```
MOV   AX, 0F00H
ADD   AX, 0200H
MOV   DX, 0000H
MOV   CX, 0010H
DIV   CX
```